CCC-C-458B
October 15, 1970
SUPERSEDING
Fed. Spec. CCC-C-458A
February 20, 1965

FEDERAL SPECIFICATION

CLOTH, FLANNEL, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton flannels (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The finished cloth shall be of the following types and classes, as specified (see 6.2):

Type I - Plain weave (soft filled sheeting), unbleached, napped one side.

Class 1 - 4.7 ounces.

Type II - Plain weave (soft filled sheeting), unbleached, napped both sides.

Class 1 - 4.6 ounces.

Type III - Plain weave (outing flannel), bleached or dyed, napped both sides (with the exception of class 4 which is dyed and napped on one side).

Class 1 - 3.5 ounces.
Class 2 - 4.0 ounces.
Class 3 - 4.6 ounces.
Class 4 - 6.0 ounces.

FSC 8305

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Type IV - Twill weave (canton flannel), unbleached, napped on the back side.

Class 1 - 5.0 ounces.
Class 2 - 6.0 ounces.
Class 3 - 7.0 ounces.
Class 4 - 8.0 ounces.
Class 5 - 10.0 ounces.
Class 6 - 12.0 ounces.

Type V - Plain weave (interlining) bleached or unbleached, napped one side.

Class 1 - 4.8 ounces.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall be equal to or better than the standard sample (see 6.3) with respect to all characteristics for which the standard sample is referenced.

3.2 Yarn. The yarn shall be made from cotton which has been carded, drawn, and spun into single yarn for both the warp and the filling.

3.3 Color. The types I, II and IV cloth shall be unbleached. The type III cloth (with the exception of class 4) shall be bleached white or dyed as specified (see 6.2). The type III, class 4 cloth shall be dyed black or any other suitable color as specified. The type V cloth shall be bleached or unbleached. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When no standard sample

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is available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.3.1 Matching. The finished type III dyed cloth shall match the standard sample under natural (north sky) daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness. The dyed and finished cloth shall show colorfastness to laundering, crocking, and perspiration equal to or better than the standard sample. When no standard sample is established, the dyed and finished cloth shall show "good" colorfastness to laundering and perspiration and shall show a Munsell Value for crocking not lower than 8.5. Tests shall be as specified in 4.4.

3.4 Physical requirement. The finished cloth shall conform to the requirements specified in tables I, II, III and IV when tested as specified in 4.4.

TABLE I. Gray soft-filled sheeting flannel

Type	Class	Weight oz./sq.yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
I	1	4.7	42	40	26	21 S/N 1/
II	1	4.6	42	40	26	19 D/N 2/

1/ Single napped (one side).

2/ Double napped (two sides).

TABLE II. Outing flannel

Type	Class	Weight oz./sq.yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
III	1	3.5	42	40	20	17 D/N 2/
III	2	4.0	42	40	23	19 D/N 2/
III	3	4.6	50	40	28	23 D/N 2/
III	4	6.0	44	40	60	35 S/N 1/

1/ Single napped (one side).

2/ Double napped (two sides).

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TABLE III. Canton flannel

Type	Class	Weight oz,/sq.yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds	
			Warp	Filling	Warp	Filling
IV	1	5.0	62	38	45	20
IV	2	6.0	62	38	60	25
IV	3	7.0	62	38	60	30
IV	4	8.0	62	38	65	40
IV	5	10.0	60	34	100	50
IV	6	12.0	60	34	100	70

TABLE IV. Interlining flannel

Type	Class	Weight oz./sq.yd. minimum	Yarns per inch minimum		Breaking strength minimum pounds		Stiffness load-pounds minimum	
			Warp	Filling	Warp	Filling	Warp	Filling
V	1	4.8	44	38	50	30	0.010	0.008

3.4.1 Weave. The weave for types I, II, III and V shall be plain. The weave for type IV shall be a 3 or 4 harness warp face twill.

3.4.2 Width. The width shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvages.

3.5 Finish. The nap of the finished cloth shall be equal to or better than the standard sample (see 3.1 and 6.3).

3.5.1 Types.

3.5.1.1 Type I cloth shall be napped one side.

3.5.1.2 Type II cloth shall be napped both sides.

3.5.1.3 Type III, classes 1, 2 and 3 cloth shall be napped both sides and the type III, class 4 cloth shall be napped on one side.

3.5.1.4 Type IV cloth shall be napped on the back side.

3.5.1.5 Type V cloth shall be napped on one side.

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3.5.2 Singeing. When specified (see 6.2), type IV, class 3 cloth shall be singed on the unnapped side.

3.6 Shrinkage. When specified (see 6.2), type I, type III, class 3 and type V cloth shall be preshrunk and the residual shrinkage shall not exceed 2.0 percent in either the warp or the filling direction when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.7 Nonfibrous material, type V. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 8.5 percent when tested as specified in 4.4.

3.8 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces as specified in PPP-P-1134.

3.10 Fiber identification. Each piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The demerit points per 100 square yards shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

4.2.2.1 Yard-by-yard examination. Each piece in the sample shall be examined on **one napped side only. When the total yardage in the** piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of **Fed. Std. No. 4, which are clearly noticeable at normal inspection distance** (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one piece within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (pieces) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one piece shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of pieces in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following values:

32.0 points - types I, II, IV and V - unbleached
28.0 points - types III and V - bleached
25.0 - type III - dyed

The lot shall be unacceptable if the points per 100 square yards of two or more individual pieces exceeds the following values:

48.0 points - types I, II, IV and V - unbleached
42.0 points - types III and V - bleached
38.0 points - type III - dyed

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If one piece exceeds the point level, a second sample shall be examined only for individual piece quality examination. The lot shall be unacceptable if one or more pieces in the second sample exceeds the point level. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- Three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Overall uncleanness.
Mottled, streaky, or cloudy (type III only).
Cloth not singed when specified (type IV, class 3 only).
Degree of napping not equal to, or greater than, standard sample. When no standard sample is available the degree of napping is not considered "good commercial".
Width less than specified.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Examination for shade (type III only). During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample which is off shade, shaded side to side, side to center,

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or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for identification of preshrinkage process and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134, to determine whether packaging, packing and marking complies with the section 5 requirements of this specification.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 2 yards full width of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as shown in the list below. The lot shall be unacceptable if one or more units fail to meet any requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE V. Test methods

Characteristic	Requirement paragraph	Test method
Cotton identification	3.2	1200 1/
Presence of labile sulfur	3.3	2020
Weave	3.4.1	Visual 2/
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Stiffness (type V):		
Warp	3.4	5202
Filling	3.4	5202
Colorfastness to:		
Laundering	3.3.2	5610
Crocking	3.3.2	5651
Perspiration	3.3.2	5680
Shrinkage	3.6	5556
Nonfibrous material (type V)	3.7	2611
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination shall be made on each sample unit and the result reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cotton flannels covered by this specification are primarily intended for use in the manufacture of pajamas, gloves, the back of glove, glove lining, padding for front interlining in coats and interlining for caps.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement document:

(a) Title, number, and date of this specification.
(b) Type and class of cloth required (see 1.2).
(c) Color of cloth required (see 3.3).
(d) When singeing is required for type IV, class 3 (see 3.5.2).
(e) Width of cloth required (see 3.4.2).
(f) When shrunk cloth for type I, type III class 3, and type V is required (see 3.6).
(g) When length of piece is other than specified (see 3.9).
(h) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

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User activities:

Army - ME
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Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

CCC-C-458B - CLOTH, FLANNEL, COTTON

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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